Introduction to Digital Logic System

SY. Btech_EXTC

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Insight

- Digital v/s Analog systems
- Advantages of Digital system
- Digital number systems
- Digital data transmission
- Applications of Digital system

Learning Outcomes

Analog – Digital Representation

Adv and Disadv of Analog- Digital system

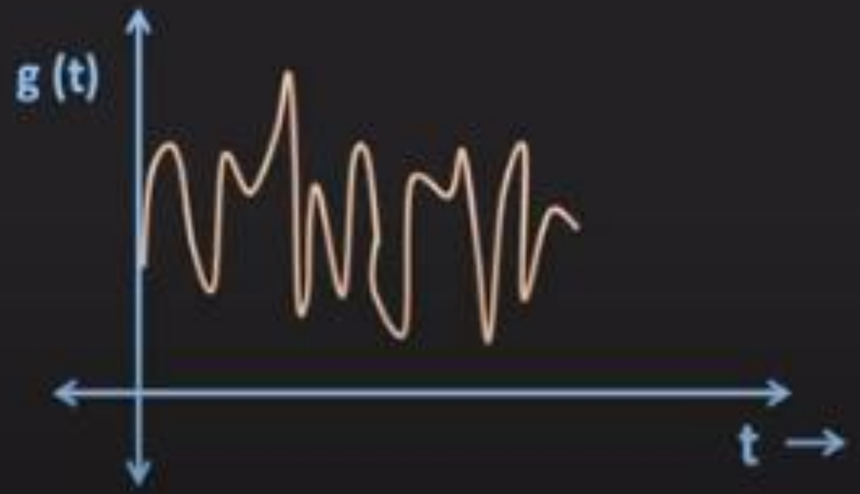
Need for digital systems

Applications

Analog vs. Digital

Analog Signal

A signal whose amplitude can take any value in the given range is the analog signal.



Analog vs. Digital

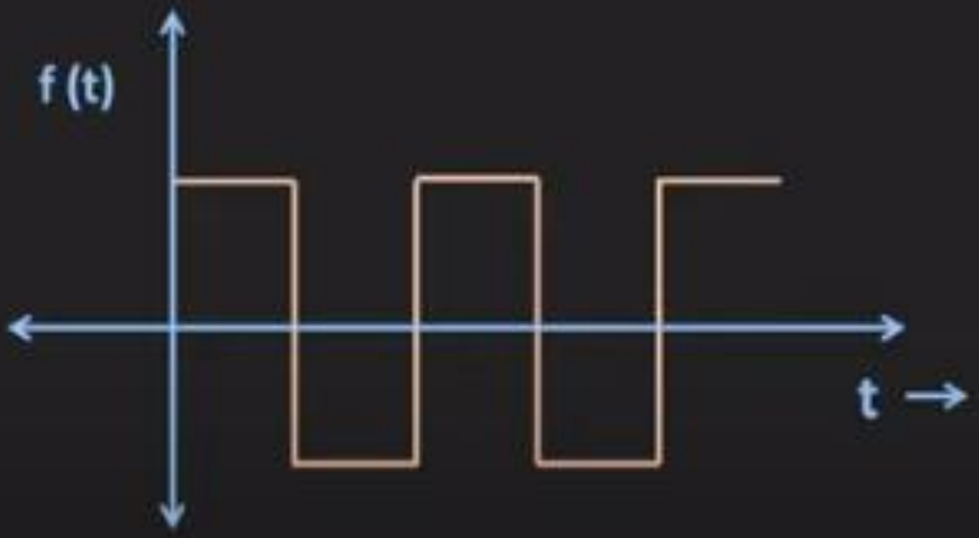
Analog Signal

A signal whose amplitude can take any value in the given range is the analog signal.

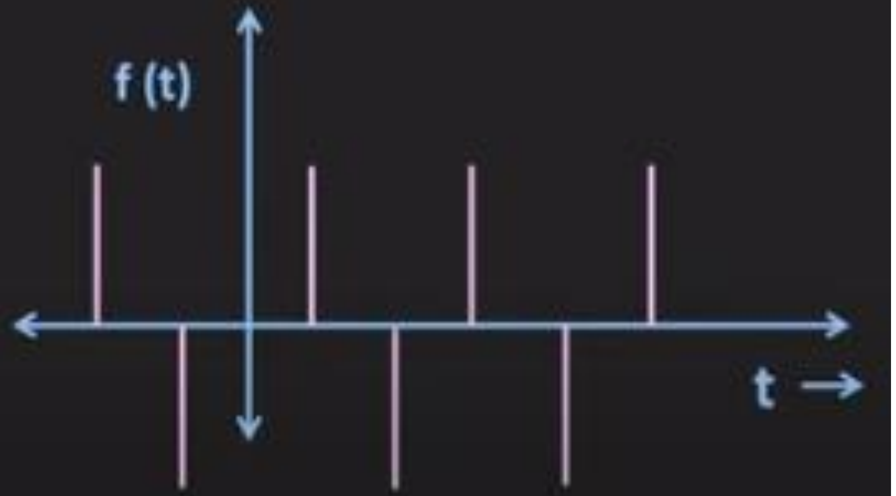
Analog vs. Digital

Digital Signal

A signal whose amplitude can take only finite number of values is the digital signal.



Digital and Continuous time signal



Digital and Discrete time signal


Digital Logic Level
High
Low
Digital Electronics
Binary Number System
High '1' → 5V
Low '0' → 0V

Numerical Representation

- Data format
  - Analog
    - 1. Speedometer
      2. Thermometer
      etc..
    - Electrical signals (V/I)
  - Digital
    - 1. Digital clock
      2. Weighing scale
      etc…

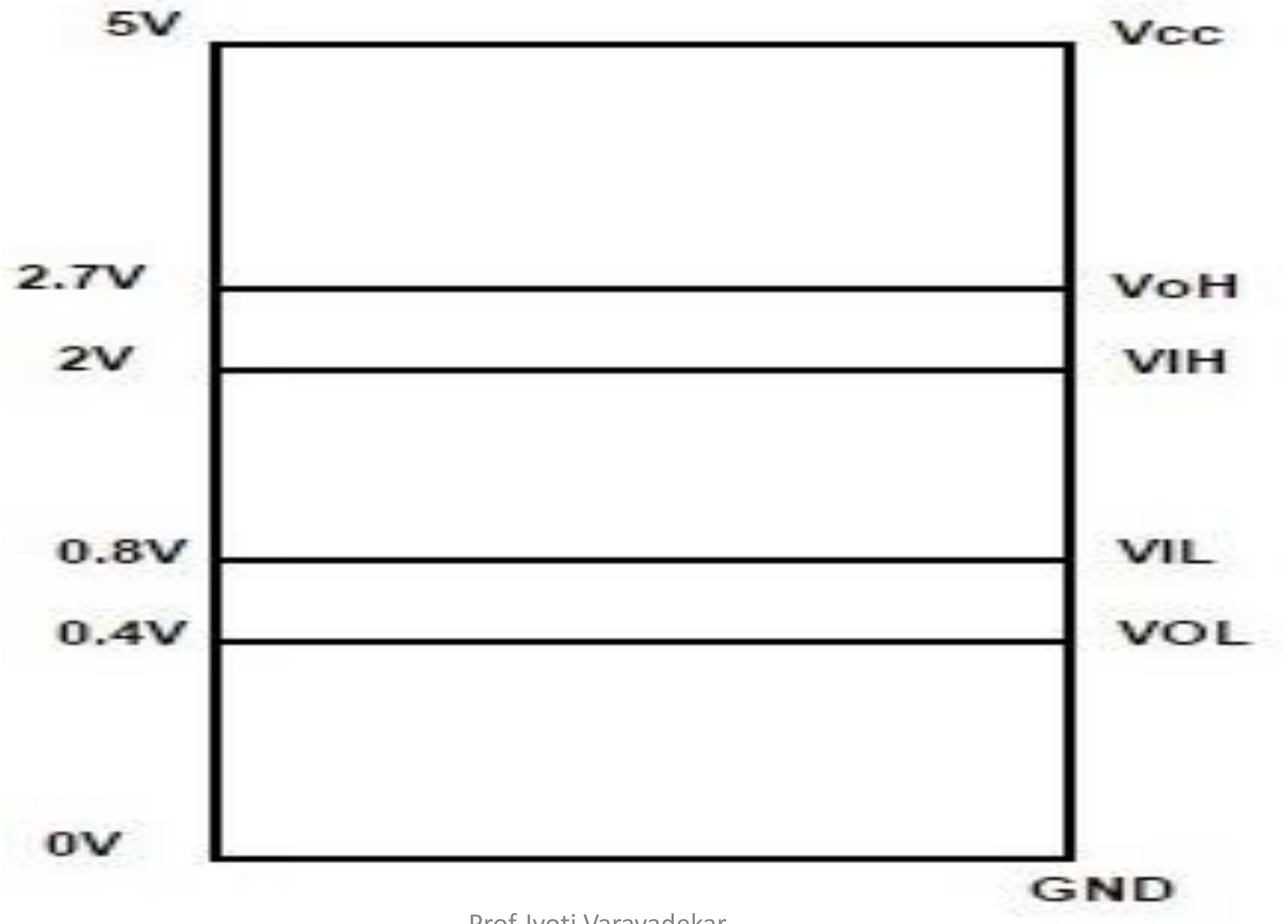
Digital v/s Analog systems

Digital systems	Analog systems
• A digital circuit consists of one of the two discrete levels/states. • Transistors used for logic gates.	• Analog circuit consists of different values w.r.t time. • Complex combination of electronic components. • Analog signals: Temperature, Pressure, Dist, Sound, V, I, P

Fundamental concepts

- Modern digital computers, calculators, communication systems, internet, email etc are operating based on the principles of digital techniques.
- These systems are *digital systems*.
- Digital systems work with *Digital signals*
- Digital signals have two discrete values or levels. Low and High
- Digital circuits are used to process these digital signals

Digital siganal level representation



In positive logic representation Bit 1 represents Logic high and Bit 0 represent a Logic low as shown in fig 2 a and b. High is represented by +5 Volts and low is represented by -5 Volts or 0 Volts.

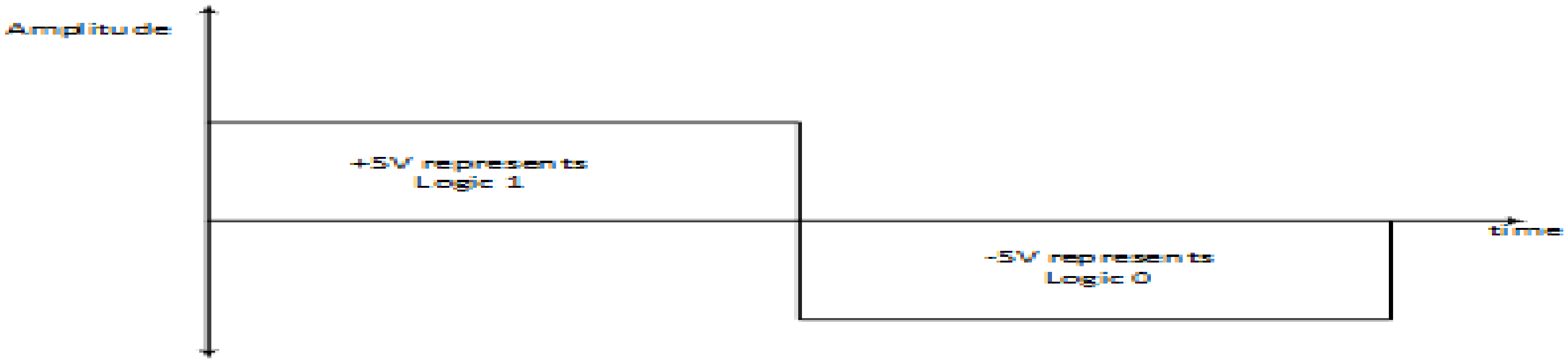


Fig. 2 a. Positive logic representation

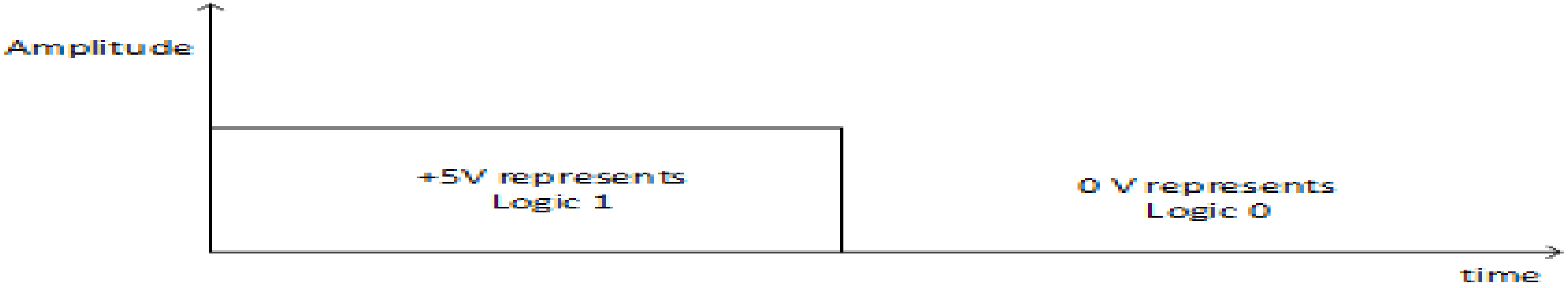


Fig. 2 b. Positive logic representation

In Negative logic representation Bit 1 represents logic low and Bit 0 represents logic high as shown in Fig 3 a and b. In terms of voltage level, bit 1 can be represented as +5V and bit 0 can be represented as 0 V or -5 Volts

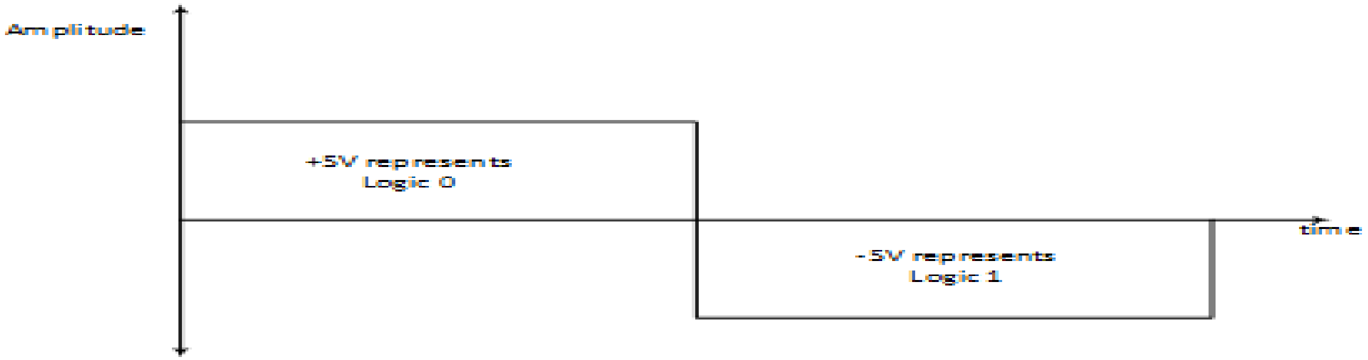


Fig. 3 a. Negative logic representation

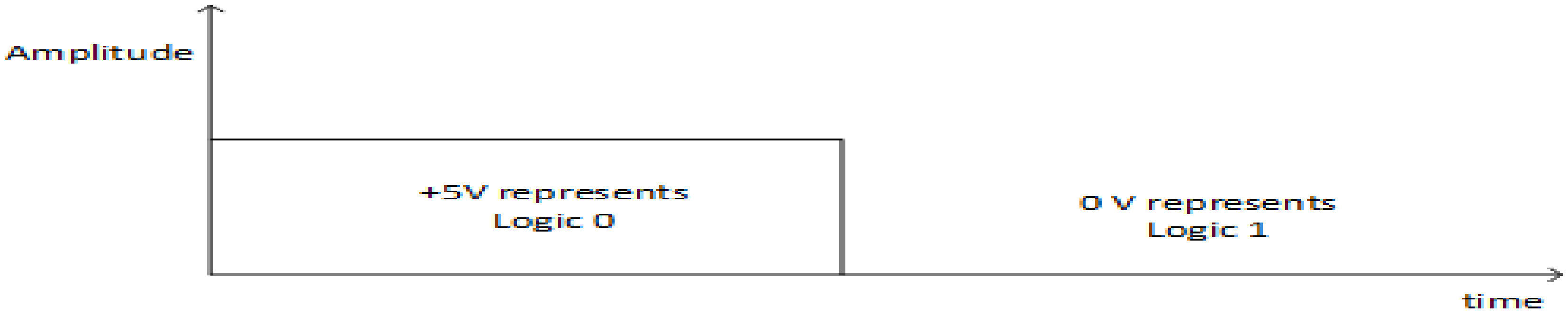
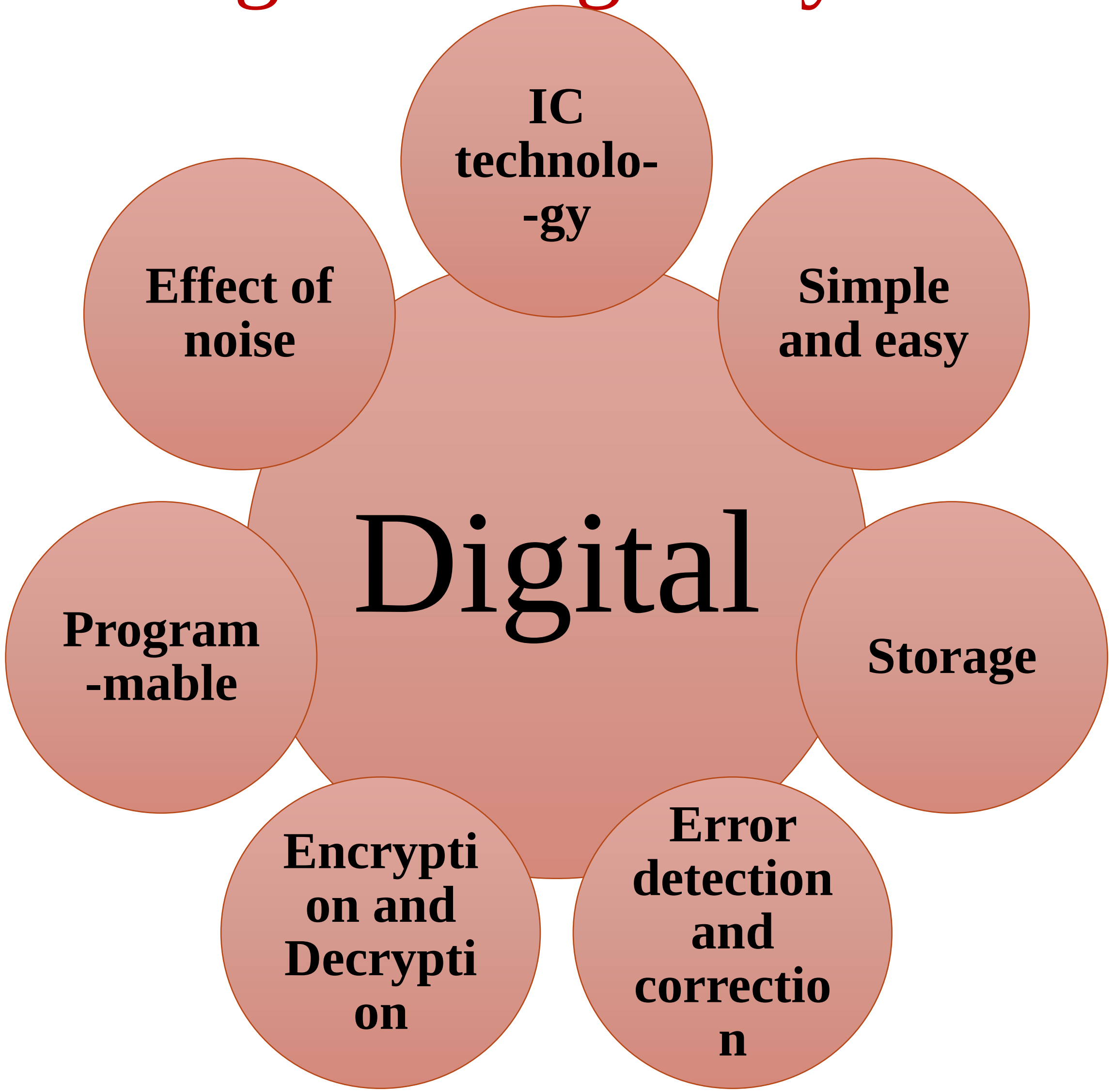


Fig. 3 b. Negative logic representation

- Two discrete signal levels can also be represented by binary digits 1 and 0.
- Binary digit is referred as a *bit*
- Binary number system is used for analysis and design of digital systems
- Two levels can be also designated as ON and OFF or TRUE and FALSE
- George Boole developed binary number systems and Boolean algebra for logic circuits design and analysis

Advantages of Digital systems



Advantages of digital signal system

- Inexpensive ICs can be used with few external components.
- Operate in one of the two **states, known as ON and OFF makes it very simple.**
- Only a few basic operations are required and are very easy to understand.
- Digital techniques deal with simple logic mathematics called Boolean algebra.

Advantages of digital signal system cond…

- Operation and network analysis of digital circuitry require simple basic concepts like switching speed and loading on the other hand, analysis of analog circuitry (Involved frequency and time domain) are quite complicated.
- Information can be stored for short periods or indefinitely.
- Data can be used for precise calculations.
- Systems can be designed more easily using compatible digital logic families.

Advantages of digital signal system cond…

- Systems can be programmed which show some manner of ‘intelligence’. A number of programmable ICs are also available.
- The display of data and other information is very convenient, accurate and elegant by using digital techniques.
- Digital circuits have capability of memory, which makes these circuits highly suitable for computers, calculators, watches, telephones, medical diagnostics, etc.
- To learn programming of digital computers it is worth knowing, the way the digital hardware works.

Disdvantages of digital signal system

- The limitations of digital circuitry are as follows:
- Most ‘real-world’ events are analog in nature.
- Analog processing is usually simpler and faster.
- Digital circuits are appearing in more and more products primarily because of
- low-cost, reliable digital ICs. Other reasons for their growing popularity are accuracy,
- added stability, computer compatibility, memory

Basic Digital Gates

- Basic operations performed in digital systems are AND, OR,NOT and FLIP-FLOP
- Digital circuits which perform these logical operations are known as Logic Gates
- Logic gates have one or more than one number of inputs but only one output.
- Basic Logic gates are : AND, OR,NOT,EX-OR,EX-NOR

Digital number system

For base or radix (r>=2) of a number system, range: zero to r-1.

Decimal Number system	Binary Number system	Octal Number system	Hexadecimal Number system
The base or radix of is 10	The base or radix of is 2	The base or radix of is 8	The base or radix of is 16
Range: 0 to 9	Range: 0 to 1	Range: 0 to 7	Range: 0 to 15

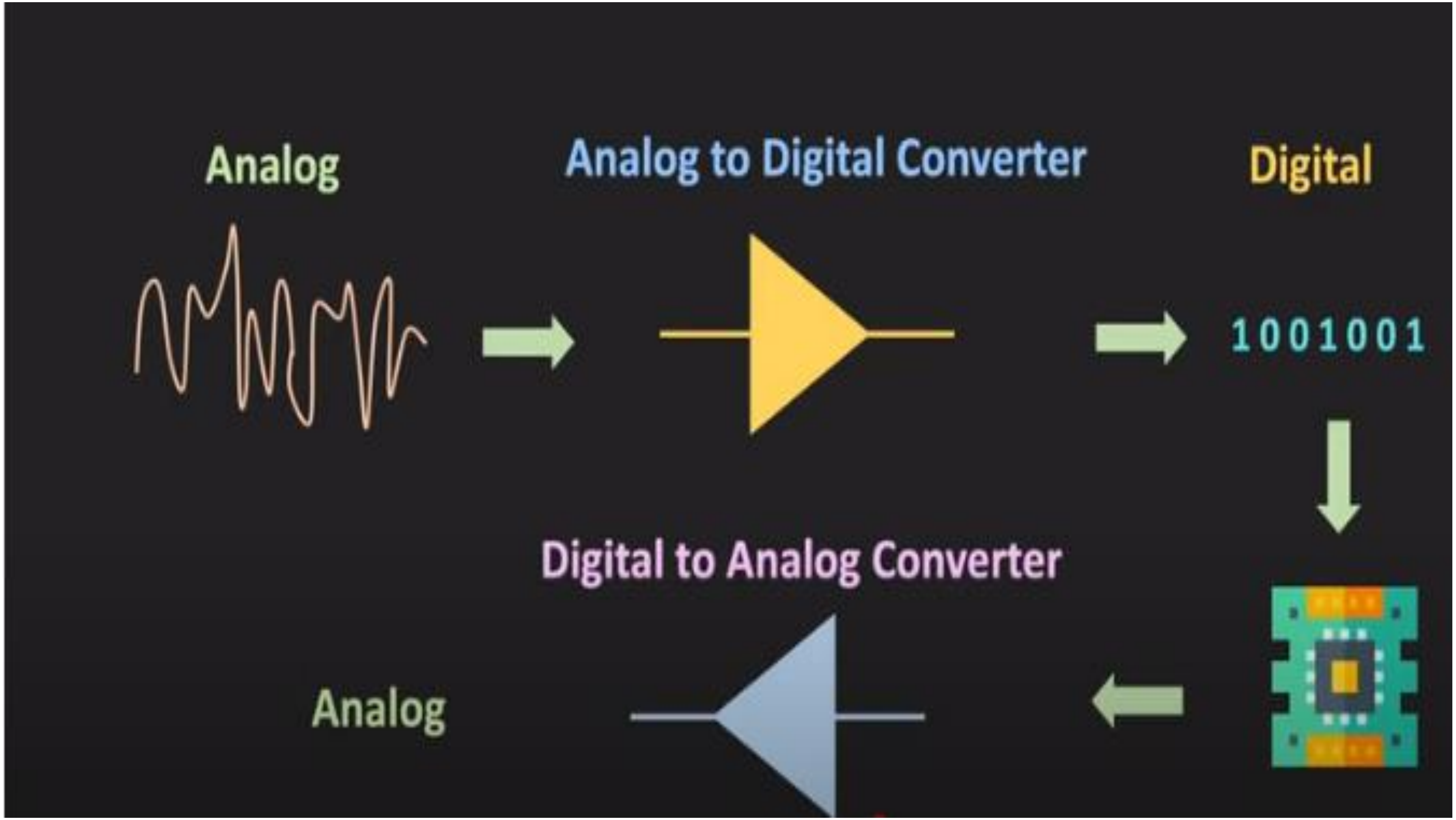
Digital data transmission

Conversion of physical variable to an electrical signal

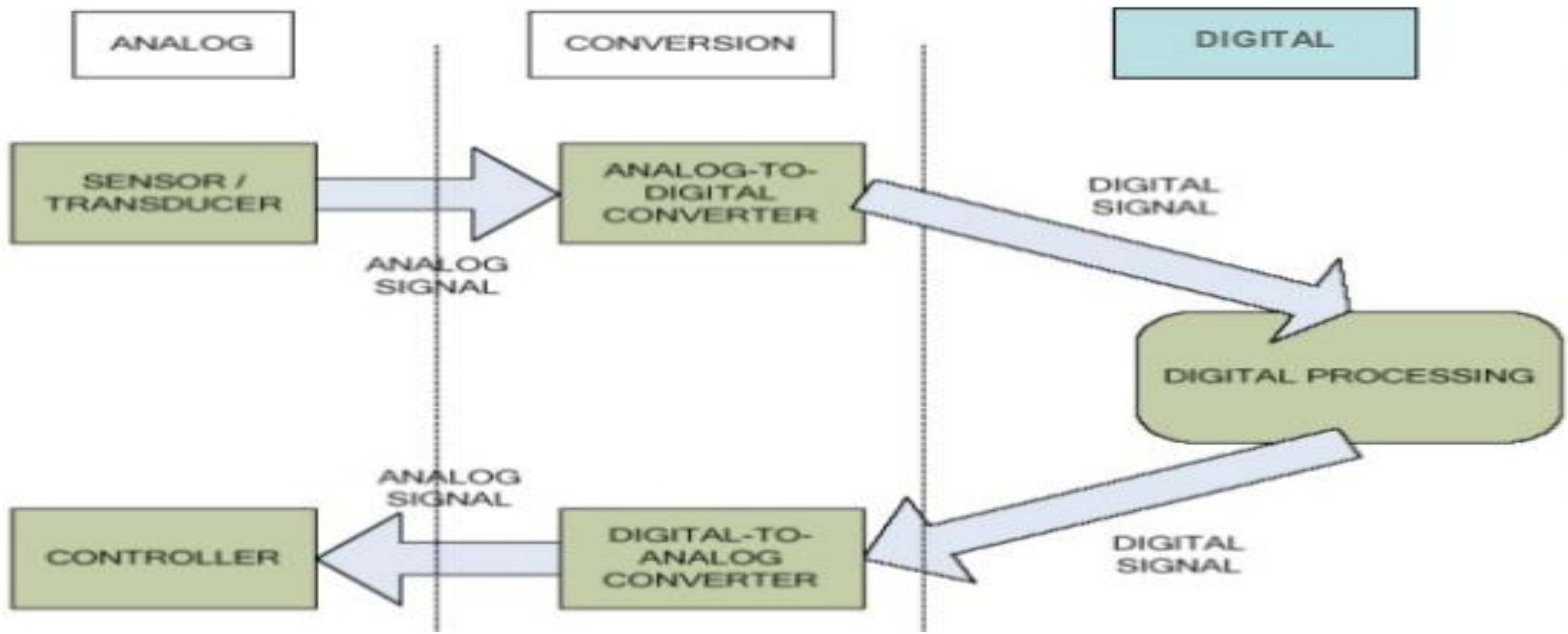
Analog to digital conversion

Processing the digital data

Digital to Analog conversion


Analog
Analog to Digital Converter
Digital
1001001
Digital to Analog Converter
Analog

Digital system (Example)



Applications of Digital system

Industrial process control

Military system

Navigation

1. Television
2. Medical equipment
3. Radar

Communication system

Thank You